

Probability Basics

Assignment



Question 1: A die is rolled. What is the probability of getting:

- (a) An even number
- (b) A number greater than 4

Question 2: In a class of 50 students:

- 20 like Mathematics (M)
- 15 like Science (S)
- 5 like both subjects

What is the probability that a student chosen at random likes Mathematics or Science?

Question 3: A bag has 3 red and 2 blue balls. If one ball is drawn randomly and is red, what is the probability that the next ball is also red (without replacement)?

Question 4: The population of a school is divided into 60% boys and 40% girls. If you want equal representation of both genders in the sample, which method should you use: Simple Random Sampling or Stratified Sampling? Why?

Question 5: The average height of 1000 students = 160 cm. A sample of 100 students shows an average height = 158 cm.

Find the **sampling error**.

Question 6: The population mean salary is ₹50,000 with $\sigma = ₹5,000$. If we take a sample of 100 employees, what is the **standard error of the mean (SEM)**?

Question 7: In a group of 100 students:

- 40 like Cricket (C)
- 30 like Football (F)
- 10 like both Cricket and Football

Find the probability that a student likes at least one sport.

Question 8: From a deck of 52 cards, two cards are drawn **without replacement**. What is the probability that both are Aces?

Question 9: A factory produces bulbs with 2% defective rate. If 5 bulbs are chosen at random, what is the probability that all are non-defective?

Question 10: Differentiate between discrete and continuous random variables with examples.